

Network Science Summer School

net-science.github.io



Universiteit Utrecht

Day program

9:00–12:00:

Introduction to network science

12:00–13:00

Lunch

13:00–16:30:

Network representation

Centrality

Intro to linear algebra

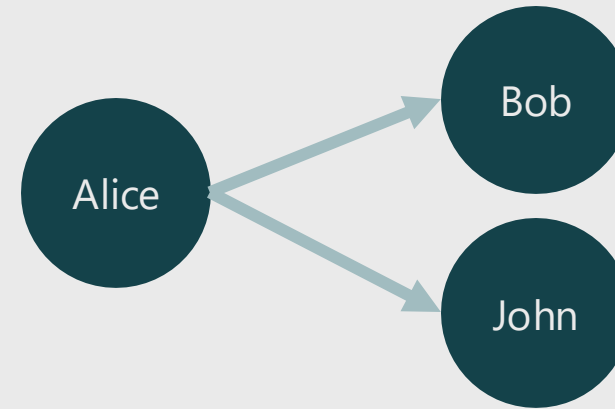
Why? Many analysis on networks rely on matrix multiplication

Network representation

Adjacency list: (edgelist)

- Adv: It is dense: Only keeping edges
- Disadvantage: Hard to work with

Source	Target	Weigth
Alice	Bob	1
Alice	John	1



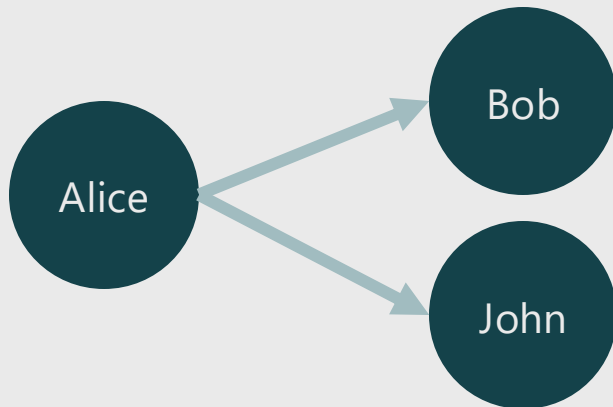
Adjacency matrix:

- Adv: Linear algebra is easy
- Disadvantage: It is sparse (mostly zeros). 1E6 nodes → 1 trillion options

Target → ↓ Source	Alice	Bob	John
Alice	0	1	1
Bob	0	0	0
John	0	0	0

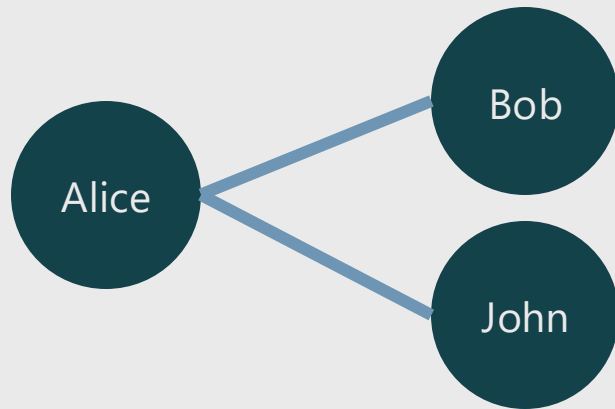
In computers → Sparse matrices: Best of both worlds

Directed networks



Target → ↓ Source	Alice	Bob	John
Alice	0	1	1
Bob	0	0	0
John	0	0	0

Undirected networks



Target → ↓ Source	Alice	Bob	John
Alice	0	1	1
Bob	1	0	0
John	1	0	0

Some terms

A =

Target → ↓ Source	Alice	Bob	John
Alice	0	1	1
Bob	0	0	0
John	0	0	0

Diagonal

Trace = Sum of elements in the diagonal

Identity matrix (I) =

$I @ A = A$

	1	0	0
	0	1	0
	0	0	1

Transpose (A^T , A') =

(python) $A.T$

Target → ↓ Source	Alice	Bob	John
Alice	0	0	0
Bob	1	0	0
John	1	0	0

Symmetric matrix: $A = A.T$ (e.g. undirected network)

Python exercise notebook 2, ex.1

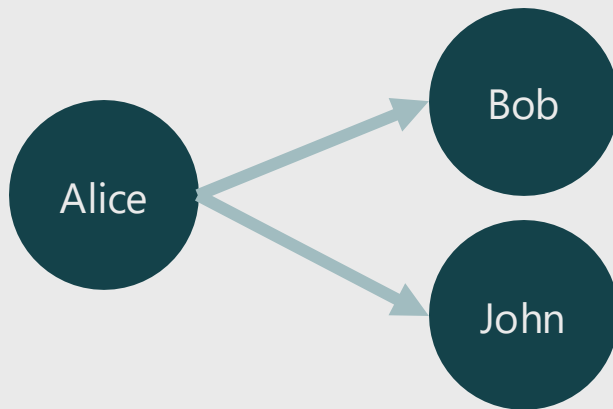
Python:

- Convert between formats
- Plot network with adjacency matrix

Transposing = reversing the edges

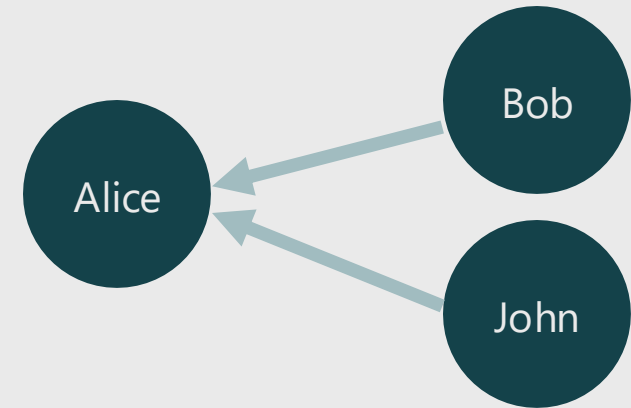
A =

Target → ↓ Source	Alice	Bob	John
Alice	0	1	1
Bob	0	0	0
John	0	0	0

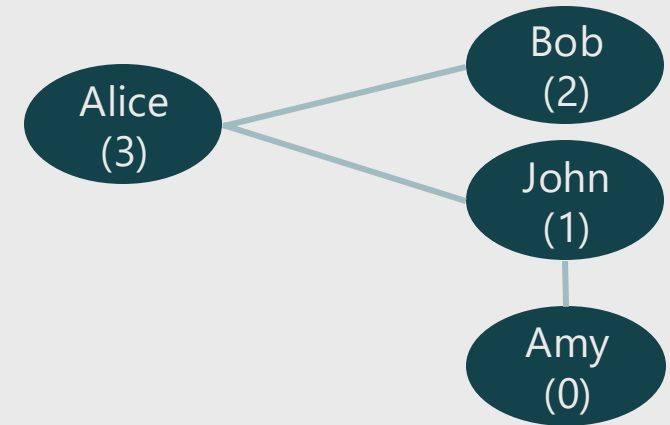


A.T =

Target → ↓ Source	Alice	Bob	John
Alice	0	0	0
Bob	1	0	0
John	1	0	0



Matrix multiplication: sum



Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	1	0	0	0
John	1	0	0	1
Amy	0	0	1	0

@

Node	children
Alice	3
Bob	2
John	1
Amy	0

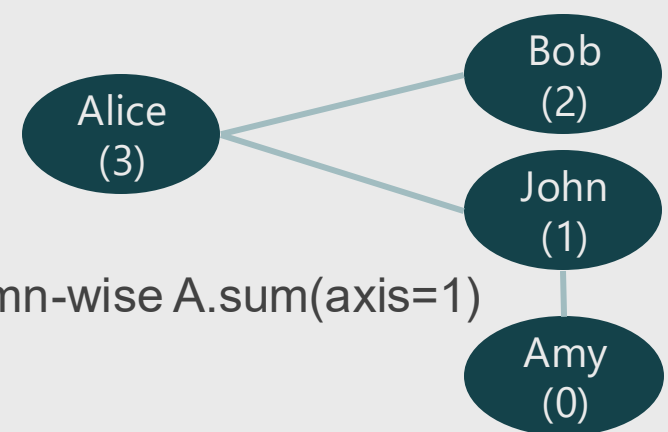
=

Node	kids
Alice	$0*3 + 1*2 + 1*1 + 0*0 = 3$
Bob	$1*3 + 0*2 + 0*1 + 0*0 = 3$
John	$1*3 + 0*2 + 0*1 + 1*0 = 3$
Amy	$0*3 + 0*2 + 1*1 + 0*0 = 1$

A @ M = SM

(N x N) @ (N x 1) = (N x 1)

Matrix multiplication: average



Divide by the degree. We get it by summing the adjacency elements column-wise `A.sum(axis=1)`

$$A @ M / A.sum(1)$$
$$(\text{N} \times \text{N}) @ (\text{N} \times \text{1}) / (\text{N} \times \text{1}) = (\text{N} \times \text{1}) / (\text{N} \times \text{1}) = (\text{N} \times \text{1})$$

Target → ↓ Origin	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	1	0	0	0
John	1	0	0	1
Amy	0	0	1	0

@

Node	Kids
Alice	3
Bob	2
John	1
Amy	0

Node	Kids
Alice	3
Bob	3
John	3
Amy	1

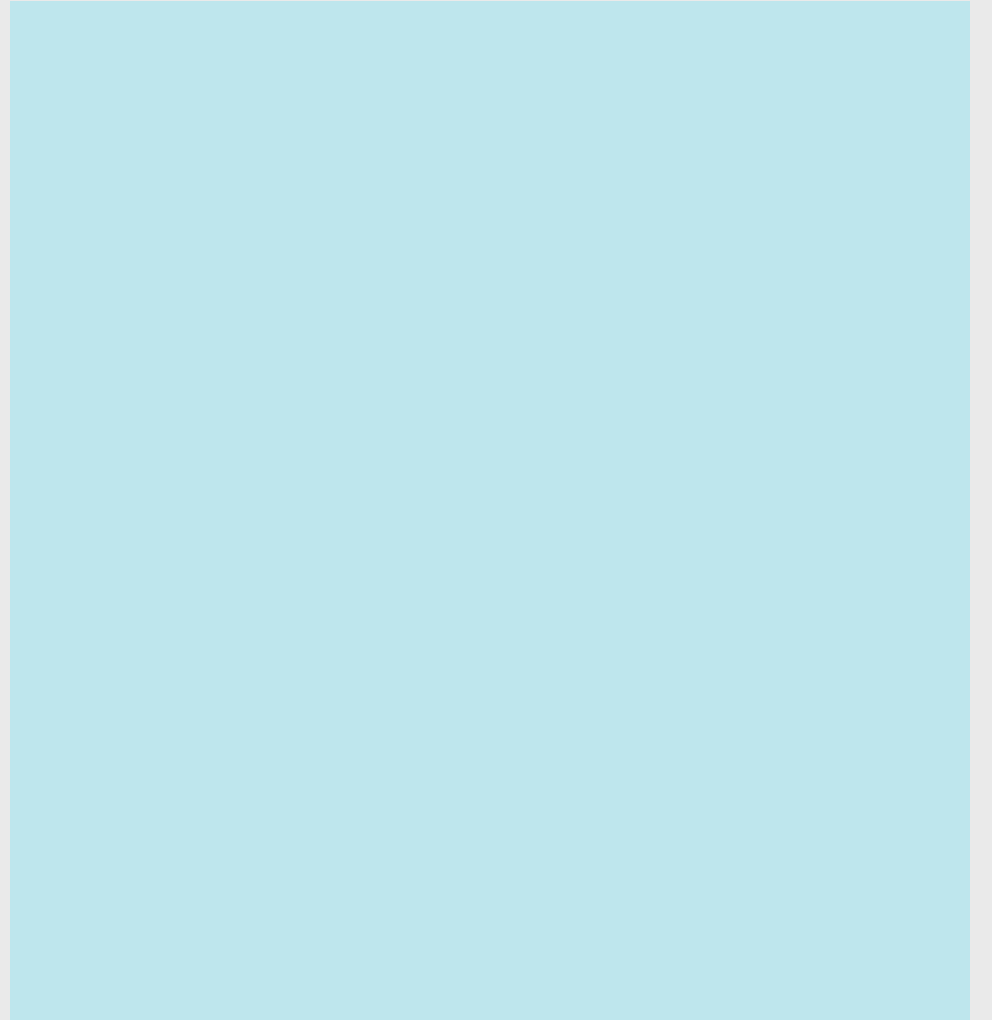
Node	Kids
Alice	1.5
Bob	3
John	1.5
Amy	1

Target → ↓ Source	Sum
Alice	2
Bob	1
John	2
Amy	1

Target → ↓ Source	Sum
Alice	2
Bob	1
John	2
Amy	1

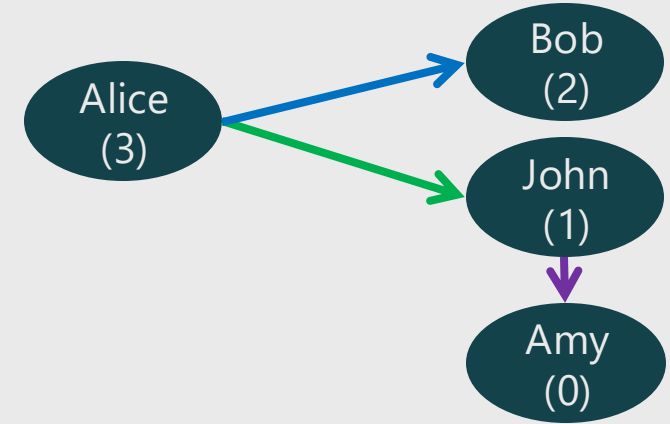
Python exercise notebook 2, ex.2

Calculate the average number of
children of your friends using matrix
multiplication



Matrix multiplication: paths

Interpretation A: Presence of path between node i and j



$A^2 =$

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

@

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

=

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0	0	1
Bob	0	0	0	0
John	0	0	0	0
Amy	0	0	0	0

From you → to your neighbors

From your neighbors → to their neighbors

You → the neig. of your neig.

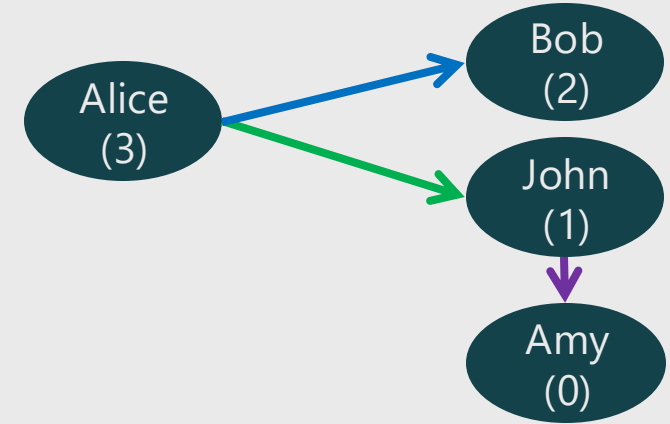
Matrix multiplication: paths

Interpretation A : Presence of path between node i and j

Interpretation A^2 : Number of path between node i and j in two steps

Interpretation A^3 : Number of path between node i and j in three steps

...



Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

@

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

=

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0	0	1
Bob	0	0	0	0
John	0	0	0	0
Amy	0	0	0	0

$$\begin{aligned} & \text{Alice} \rightarrow \text{Alice} (0) * \text{Alice} \rightarrow \text{Amy} (0) \\ & + \text{Alice} \rightarrow \text{Bob} (1) * \text{Bob} \rightarrow \text{Amy} (0) \\ & + \text{Alice} \rightarrow \text{John} (1) * \text{John} \rightarrow \text{Amy} (1) \\ & + \text{Alice} \rightarrow \text{Alice} (0) * \text{Alice} \rightarrow \text{Amy} (1) \end{aligned}$$

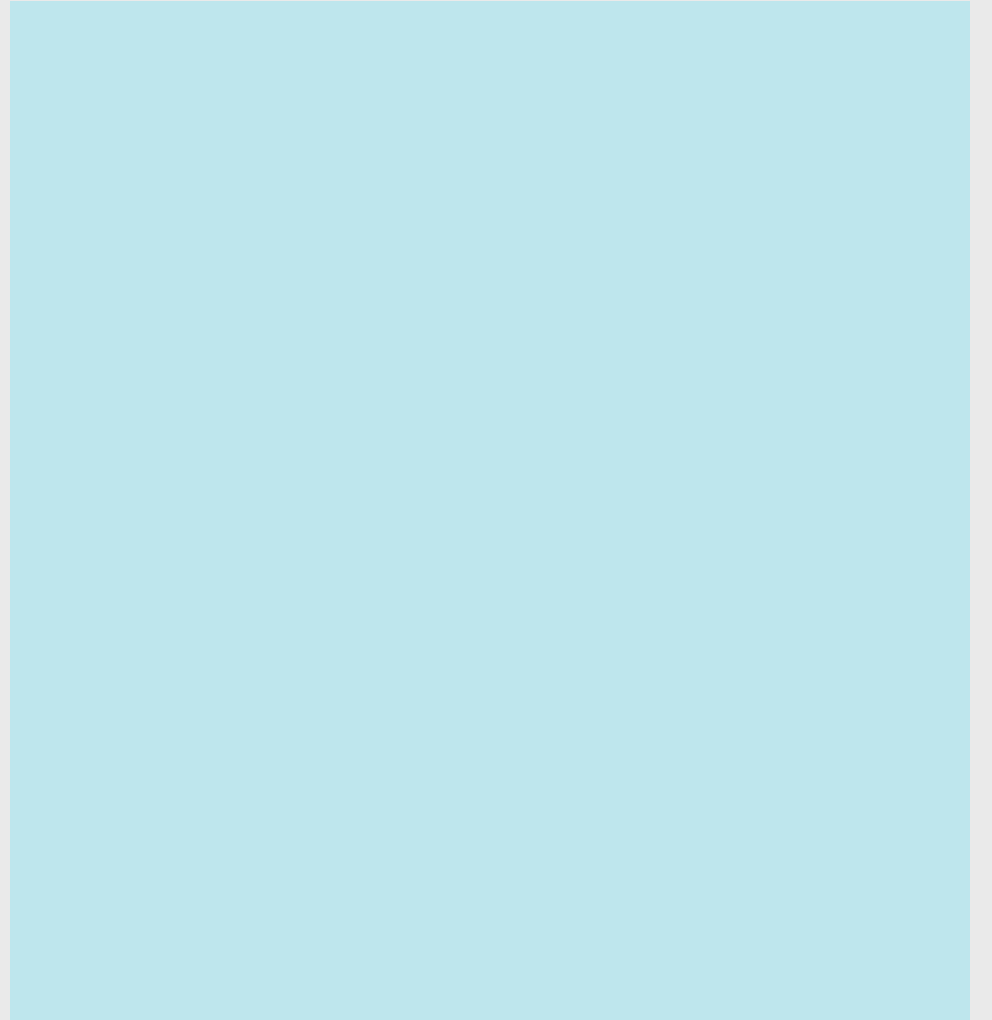
Matrix multiplication: number of people reached in <3 steps

Number of paths in two or three steps from node i to node j: $N = A + A^2 + A^3$

We need to remove duplicate paths: $N = N > 0$

We need to remove paths from us to ourselves $N.setdiag(0)$

Python exercise notebook 2, ex.3a



Matrix multiplication: number of triangles

We are interested in the diagonal of A^3

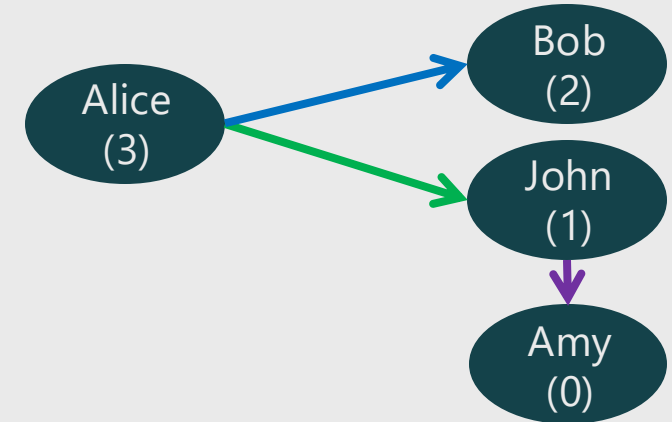
A triangle is a path in from node i to node i .

The diagonal contains the number of triangles in which node i appears.

Undirected network? Divide the triangles by two (two directions)

Counting the total number of triangles? Divide the trace by 3 (each triangle has 3 members)

Matrix multiplication: number of triangles



A^2

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0	0	1
Bob	0	0	0	0
John	0	0	0	0
Amy	0	0	0	0

@

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

=

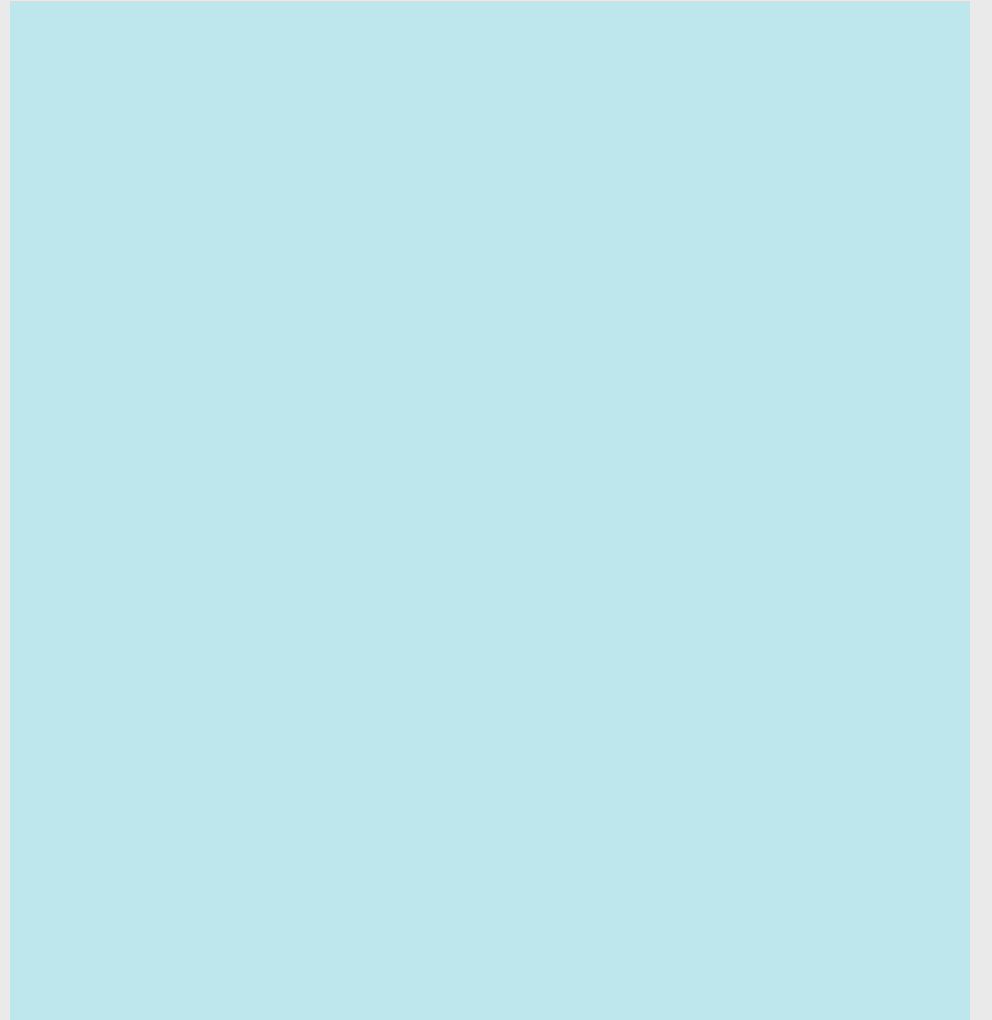
Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0	0	0
Bob	0	0	0	0
John	0	0	0	0
Amy	0	0	0	0

Alice → Alice in two steps * Alice → Alice (0)
 Alice → Bob in two steps * Bob → Alice (0)
 Alice → John in two steps * John → Alice (0)
 Alice → Amy in two steps * Amy → Alice (0)

Diagonal of A^3

Alice → X_1 * X_1 → X_1 * X_1 → Alice +
 Alice → X_1 * X_1 → X_2 * X_2 → Alice +
 ...

Python exercise notebook 2, ex.3b



Centrality measures

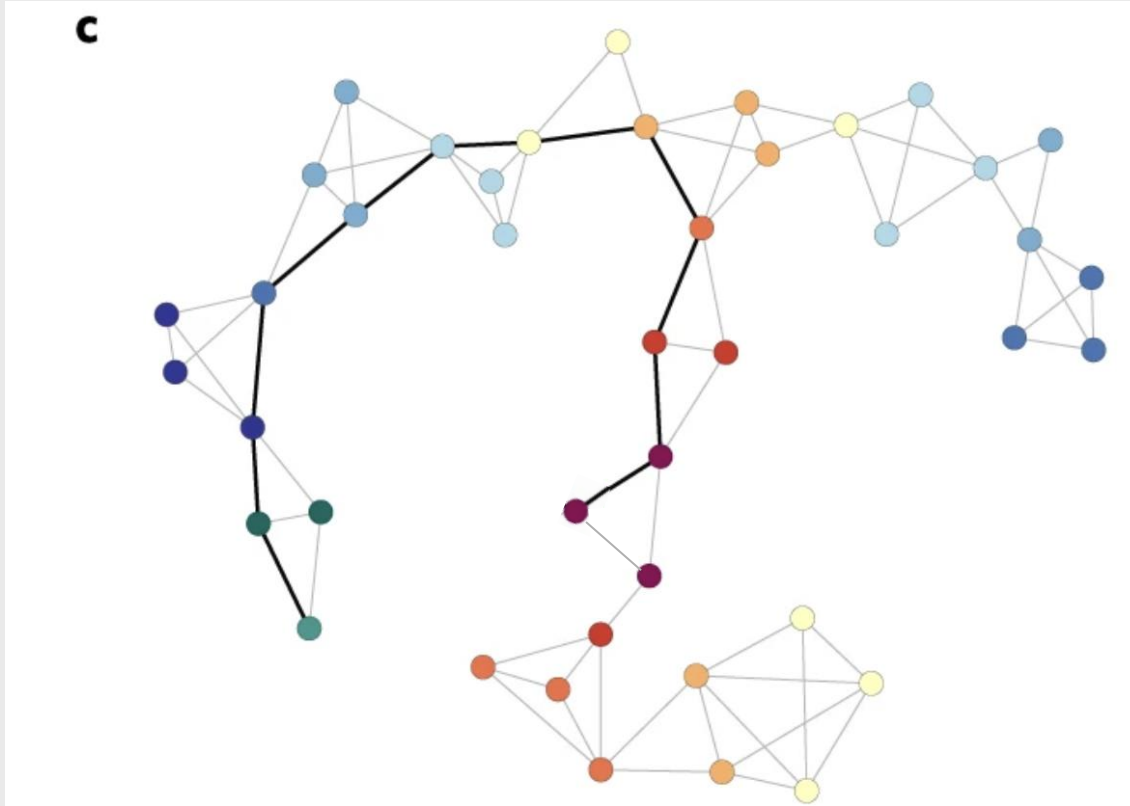
Nice explanations:

<https://aksakalli.github.io/2017/07/17/network-centrality-measures-and-their-visualization.html>

Networks: an introduction (Newman)

Motivating examples

How to stop the spread of diseases?



Important nodes: bridges

How to sort Google results?

PageRank counts the **quality** and **quantity** of backlinks to assess the importance of a page.



<https://www.leannewong.co/google-pagerank/>

Important nodes: those linked by important nodes

Centrality

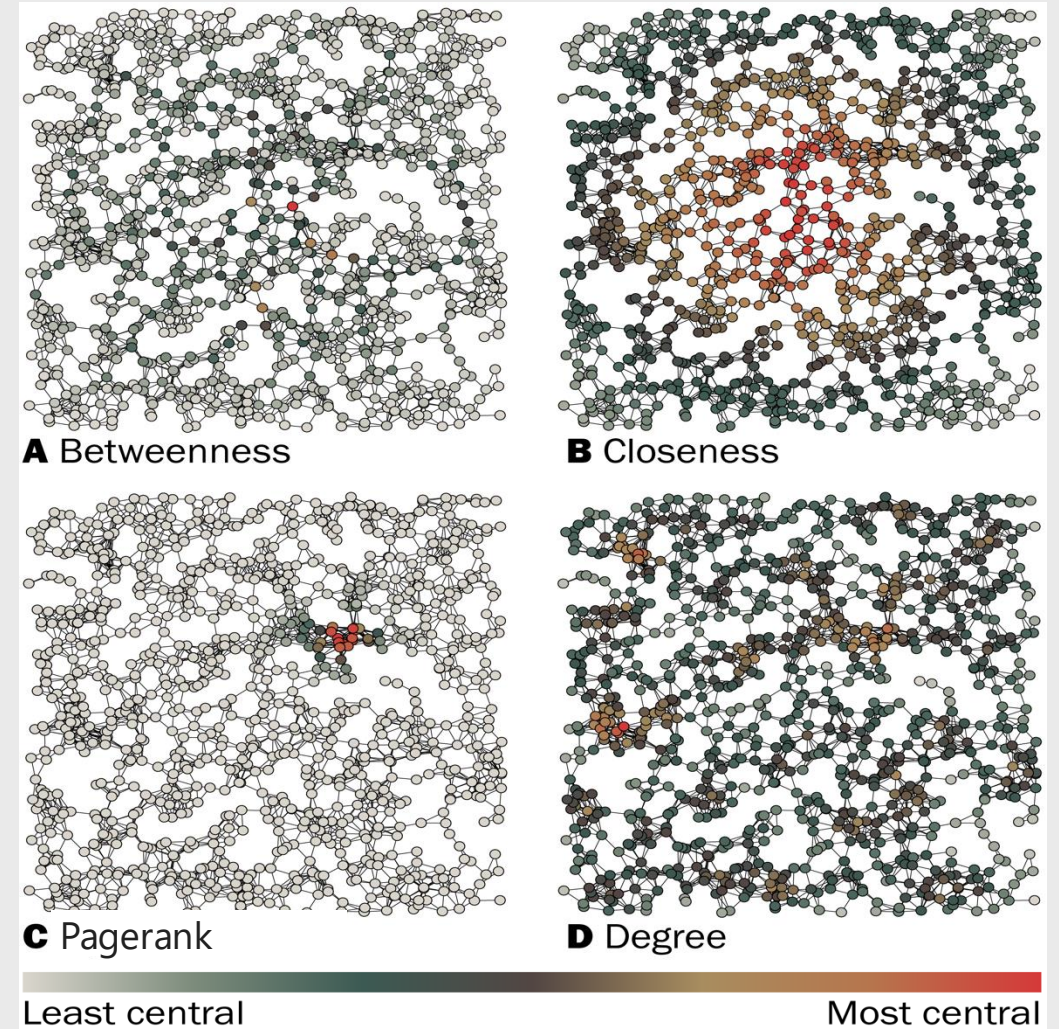
Who are the key actors in the network?

Centrality measures provide answers to this question.

Different centrality measures define importance in different ways :

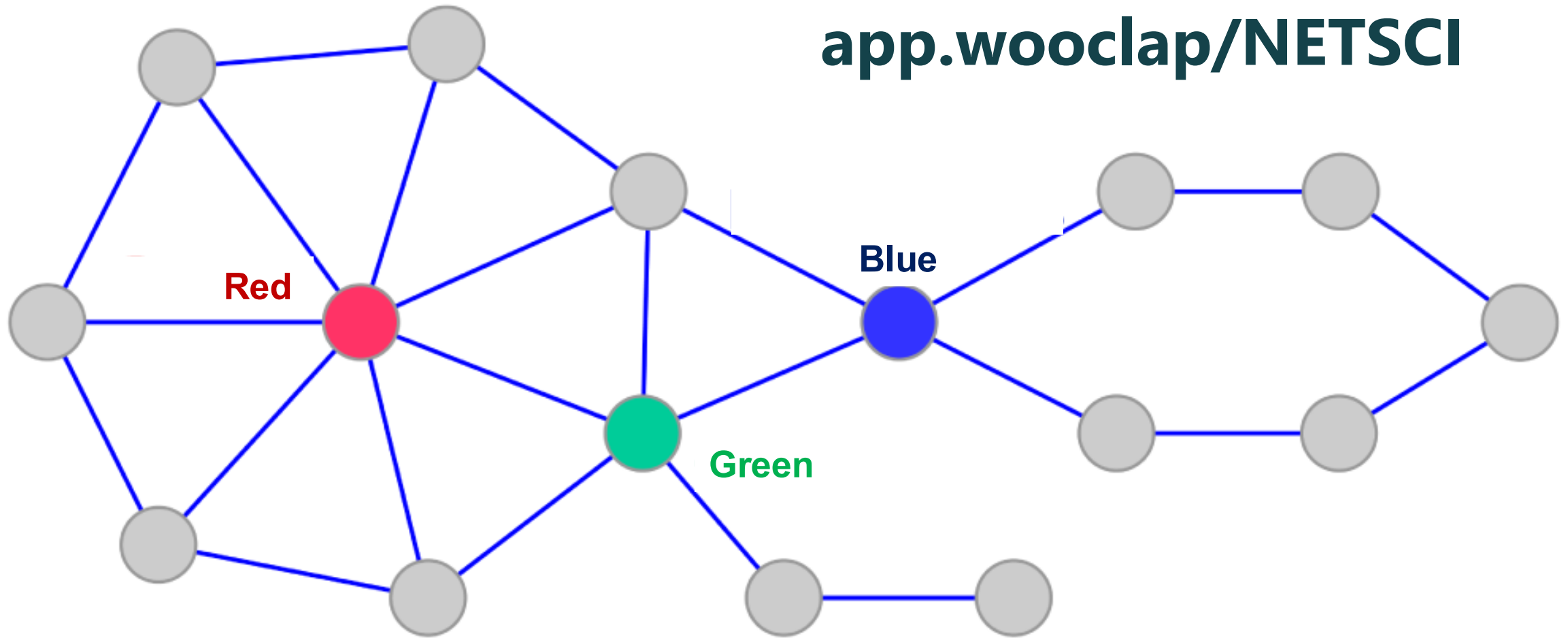
- *Degree*: Connected to many nodes
- *Closeness*: Close to all other nodes
- *Betweenness*: In the middle of shortest paths
- *Pagerank*: Connected to important nodes

Centrality identify *the most important nodes*. It does not quantify the importance of nodes in general. The relative rankings of non-important nodes may be meaningless.



Which node has higher degree/betweenness/closeness?

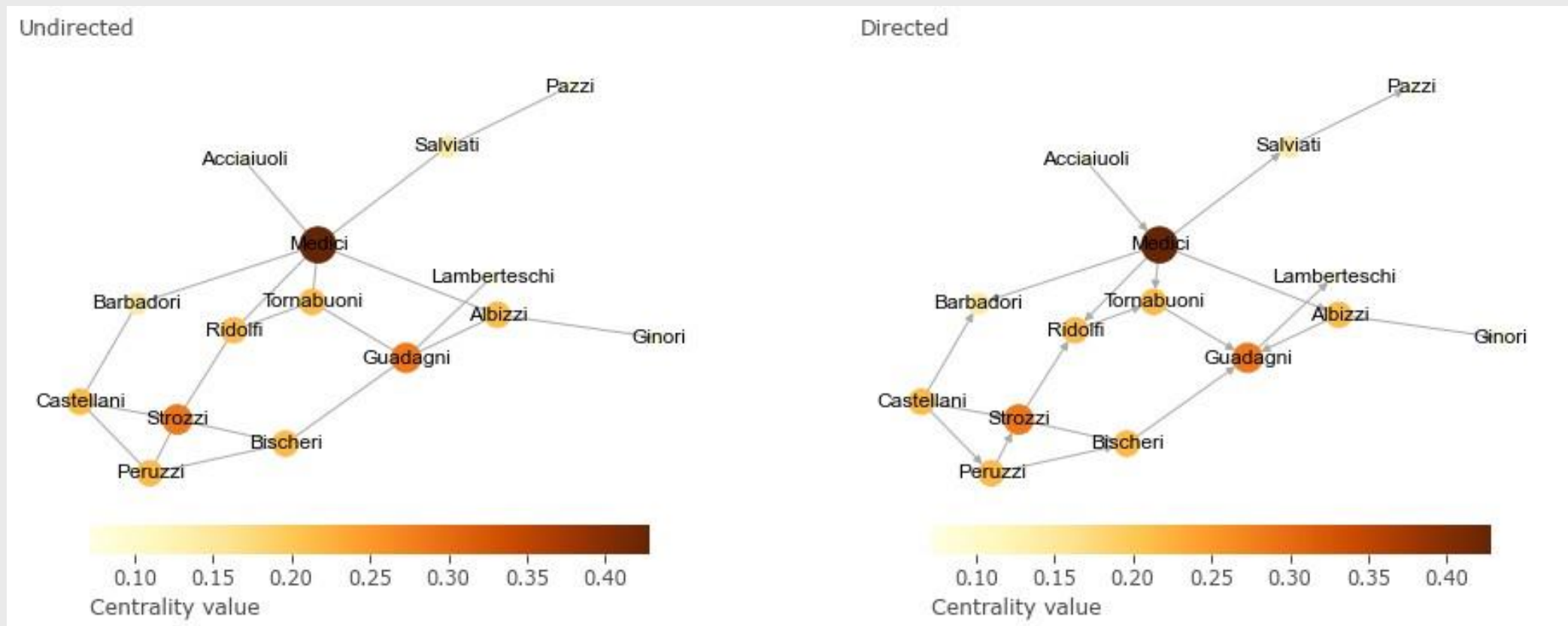
app.wooclap/NETSCI



Degree centrality: $DC_i \propto d_i$

d_i = degree of node i

Measures the **local** influence of the node



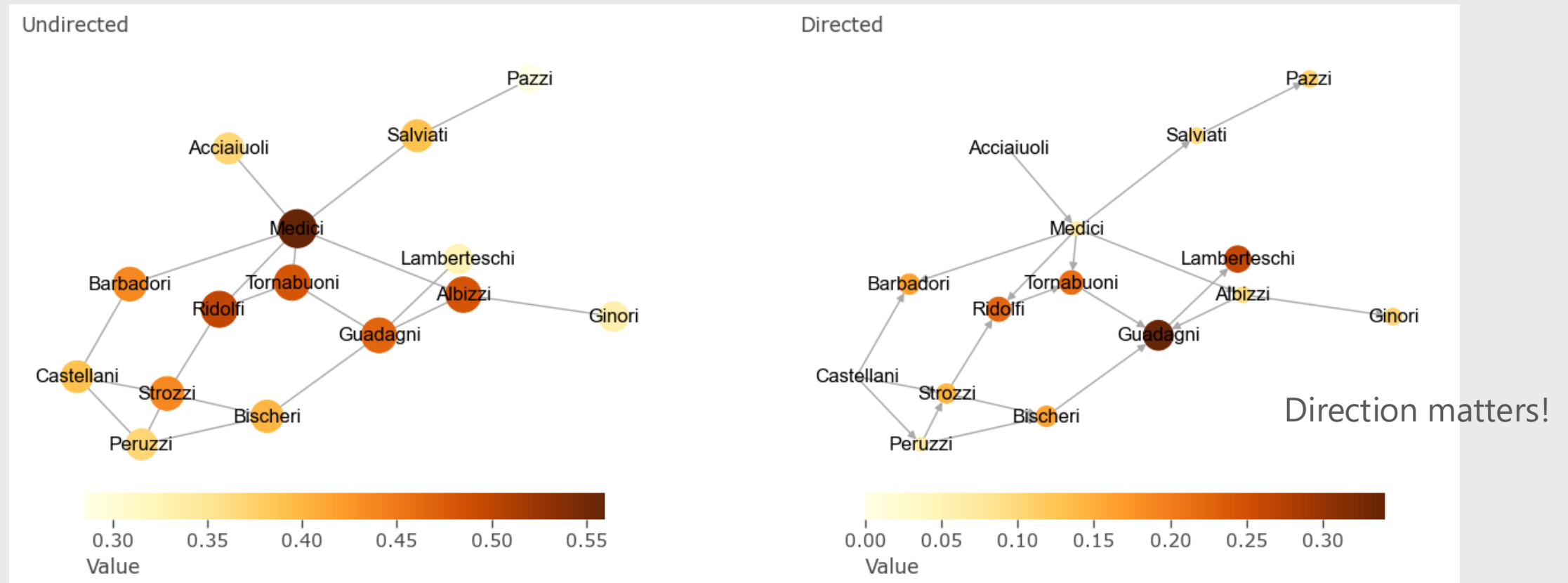
Closeness centrality: $CC_i = \frac{1}{l_i} = \frac{N}{\sum_j d_{ij}}$

l_i = average distance of node i to all other nodes

d_{ij} = shortest distance from node i to node j

Only useful in fully connected networks

Measures the **most central** node in the network (closest to get to all other nodes)

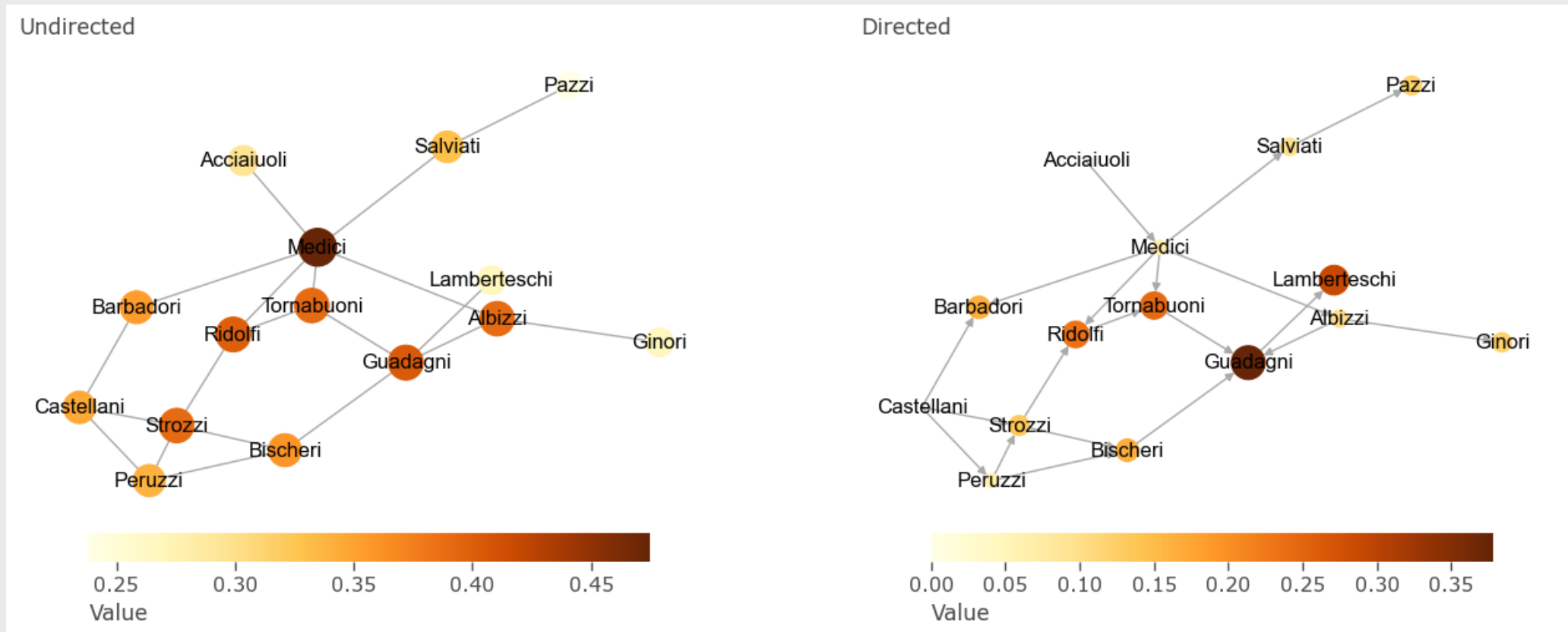


Harmonic closeness centrality: $HCC_i \propto \sum_{ij|i \neq j} 1/d_{ij}$

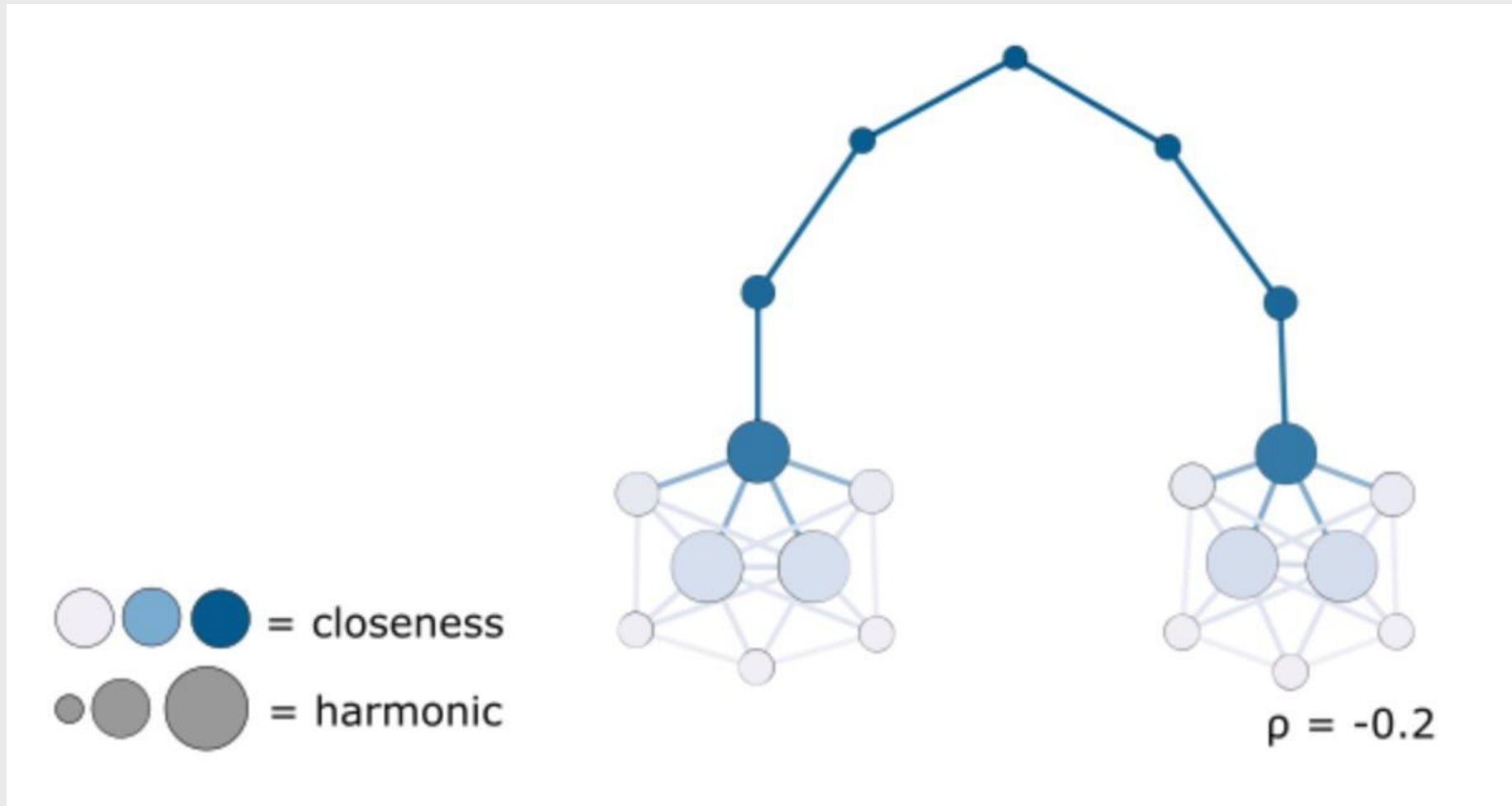
d_{ij} = shortest distance from node i to node j

Useful also in disconnected networks. Gives more weight to closer nodes.

Measures the **most central** node in the network (harmonic average)



Closeness vs harmonic closeness



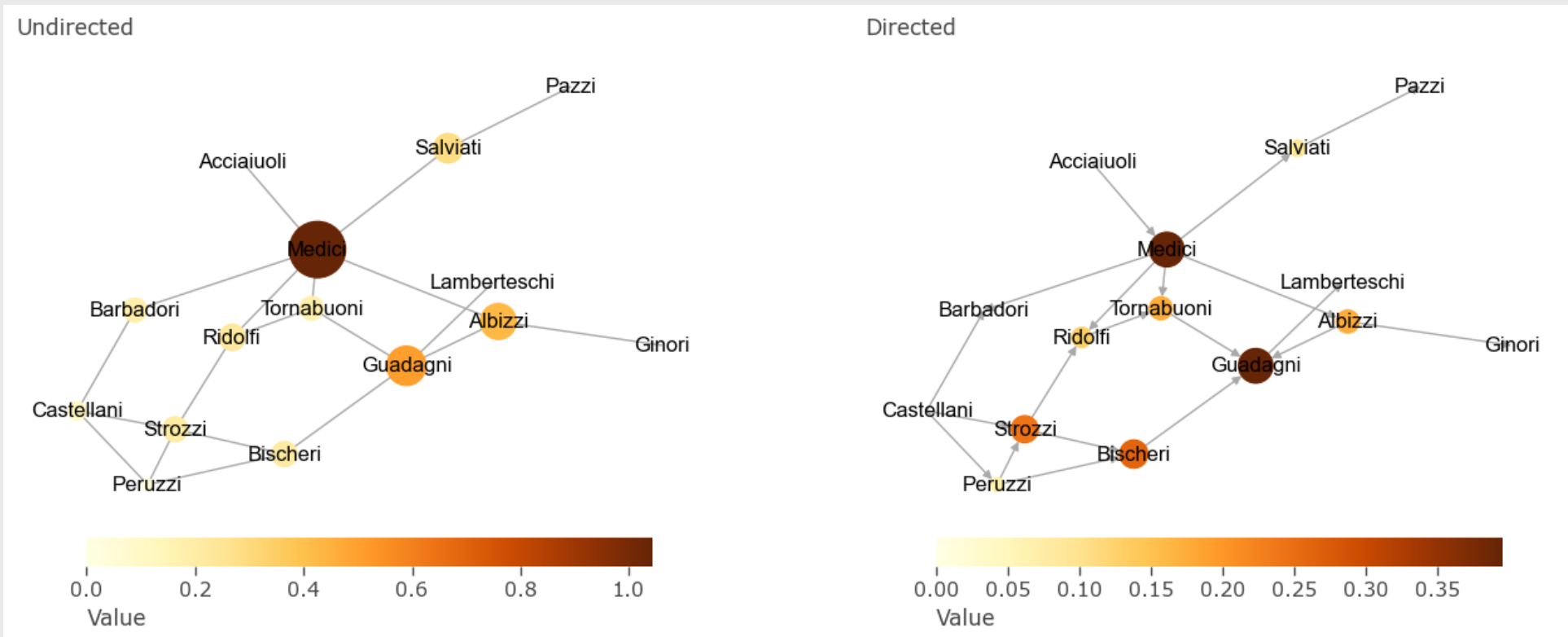
Betweenness centrality: $BC_i \propto \sum_{st} n_{st}^i$

$n_{st}^i = 1/g$ if node i lies on the g shortest paths between nodes s and t

Measures **brokerage** in the network → disruption of these nodes = disruption of communication

Assumptions:

- every pair of nodes in the network exchanges messages at the same average rate
- messages always take the shortest available path through the network



*Freeman (1977),
and Anthonisse
(1971, unpublished)*

Eigenvector centrality: $EC_i = \lambda^{-1} \sum_j A_{ij} EC_j$

Considers how central your neighbors are.

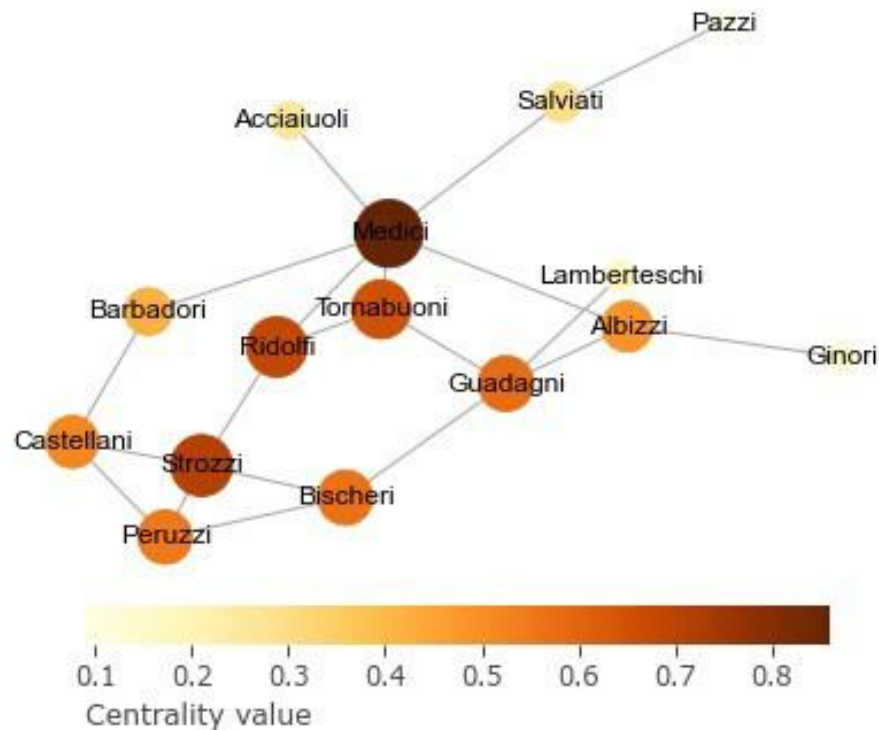
EC_j = eigenvector centrality of node j

λ = largest eigenvalue

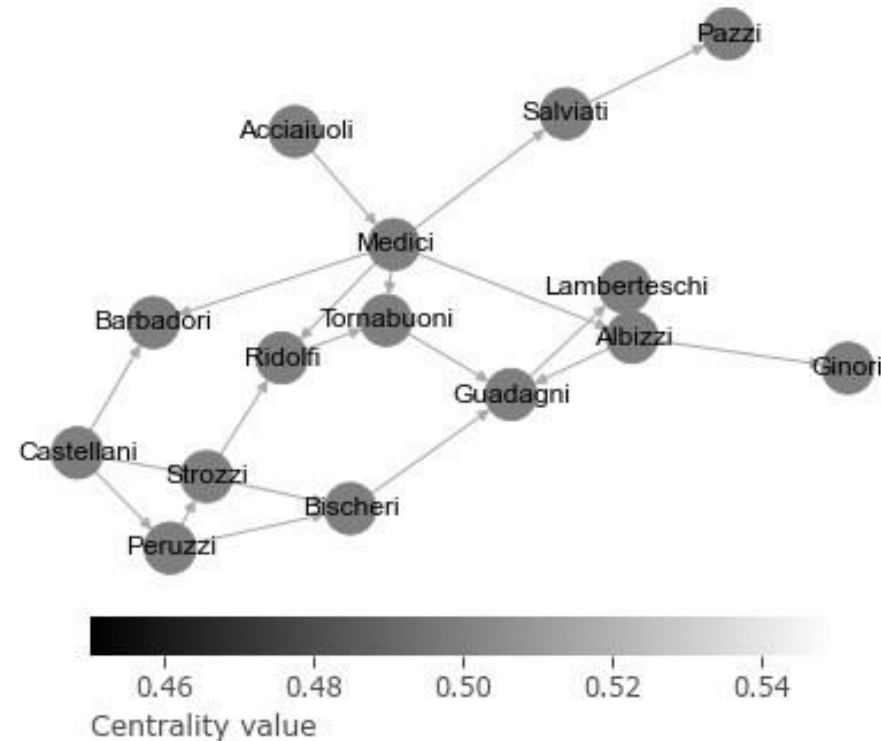
Measures total **influence** in the network (assuming all nodes are the same)

Only for undirected, fully-connected networks!

Undirected



Directed



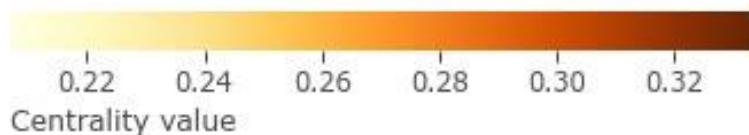
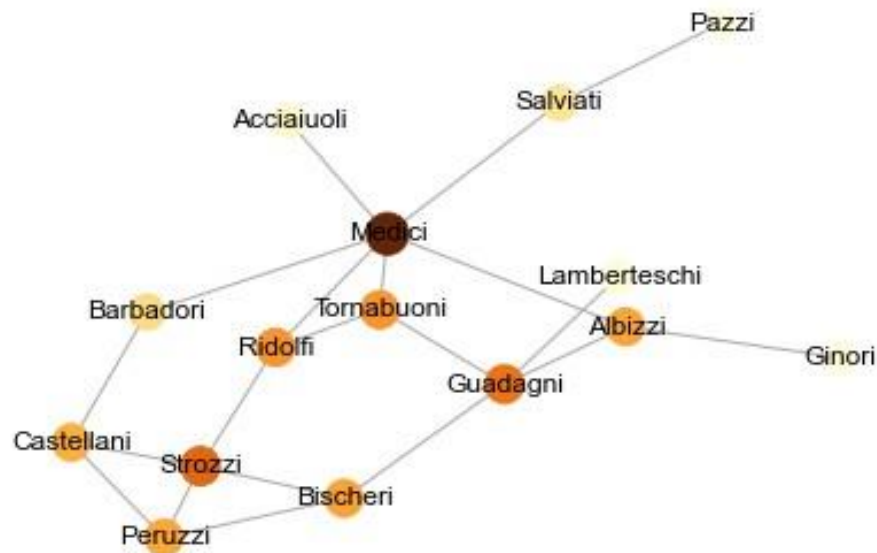
Katz centrality: $KC_i = \alpha \sum_j A_{ij} KC_j + \beta$

k_j = Katz centrality of node j

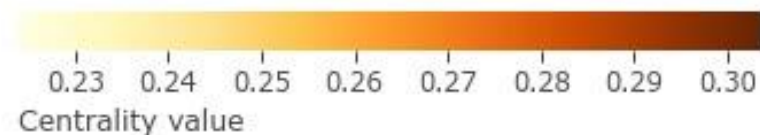
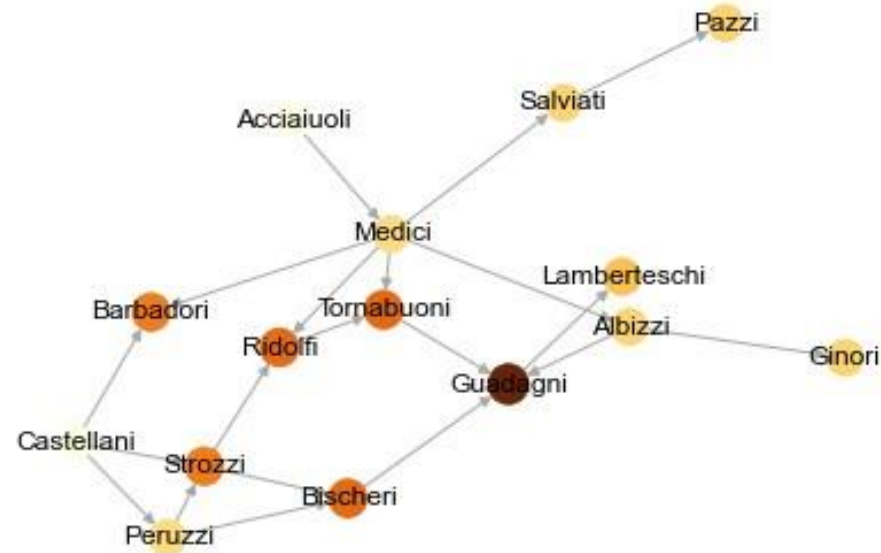
Takes into account how central your neighbors are, **each node has a minimum value of β** , and the balance between the constant and the eigenvector part is controlled by α

Measures total **influence** in the network (assuming all nodes are the same)

Undirected



Directed

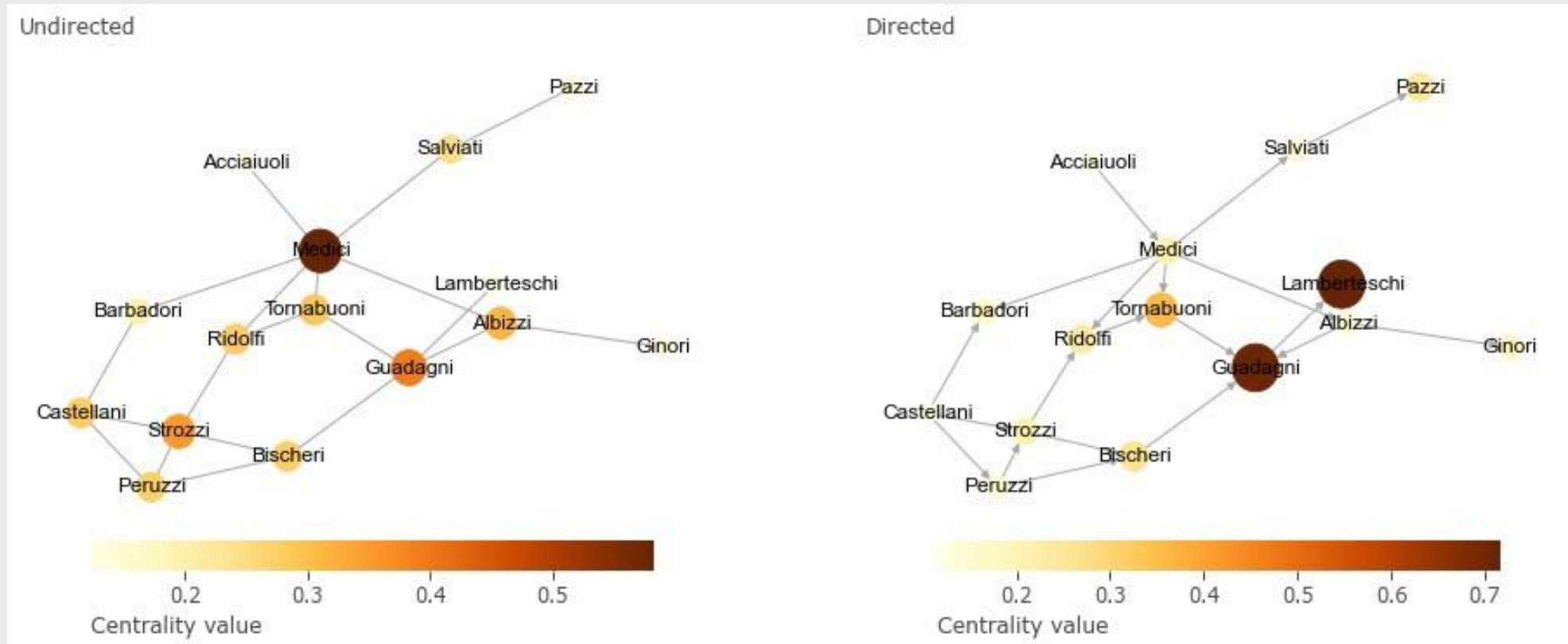


PageRank centrality: $PR_i = (1 - \alpha) \sum_j \frac{A_{ij} PR_j}{d_j} + \alpha$

d_j = Degree of node j . PR_j = Pagerank centrality of node j

Takes into account how central your neighbors are. Each node has a minimum value of α . The PageRank of a node is α plus **the PageRank of your neighbors** (normalized by their out-degree)

Measures total **influence** in the network (assuming all nodes are similar)



Bonacich, 1987

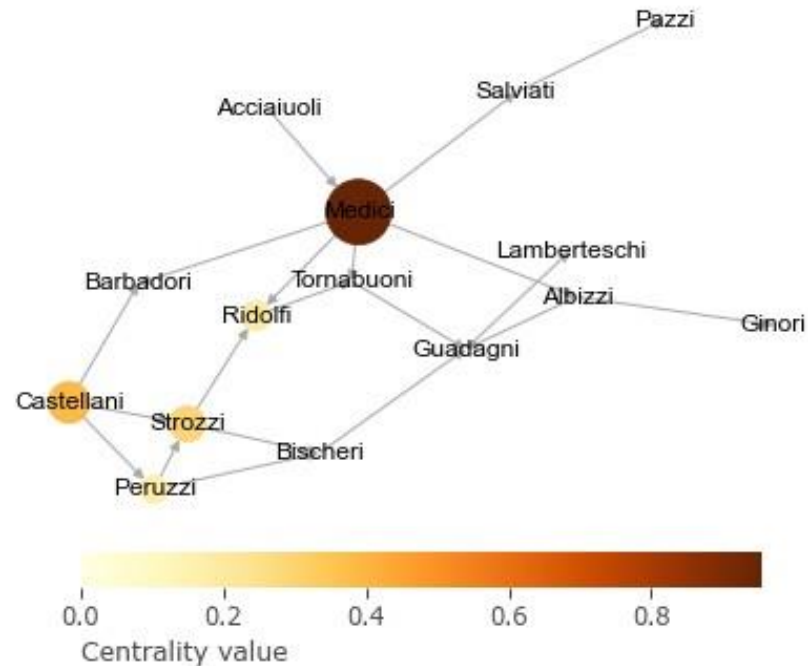
Hubs and authorities (HITS)

A node may be important if it points to others with high centrality, e.g., a review article pointing to prestigious articles

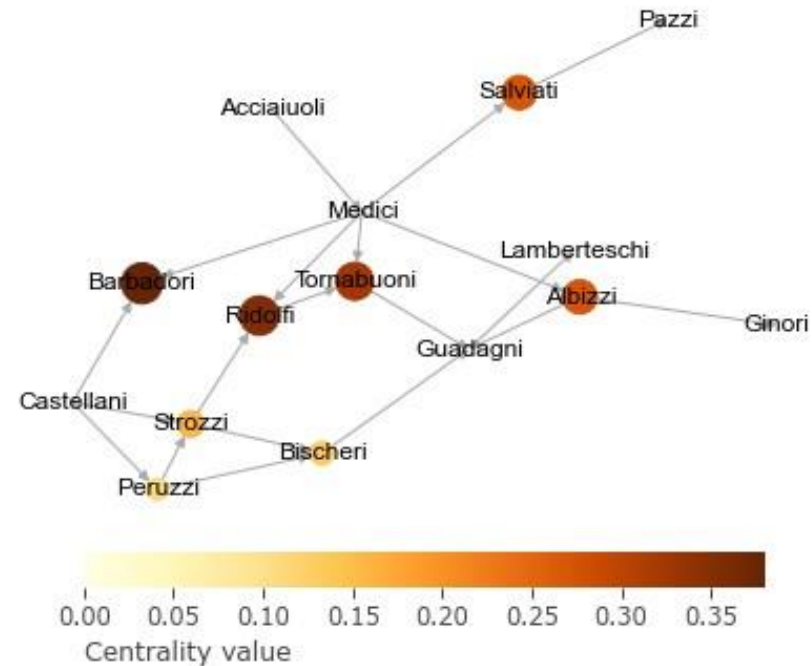
Authorities are nodes that contain useful information on a topic of interest and **hubs** are nodes that tell us where the best authorities are to be found (Newman). Two centralities: authority (a) and hub (h) centrality. **Only for directed networks!**

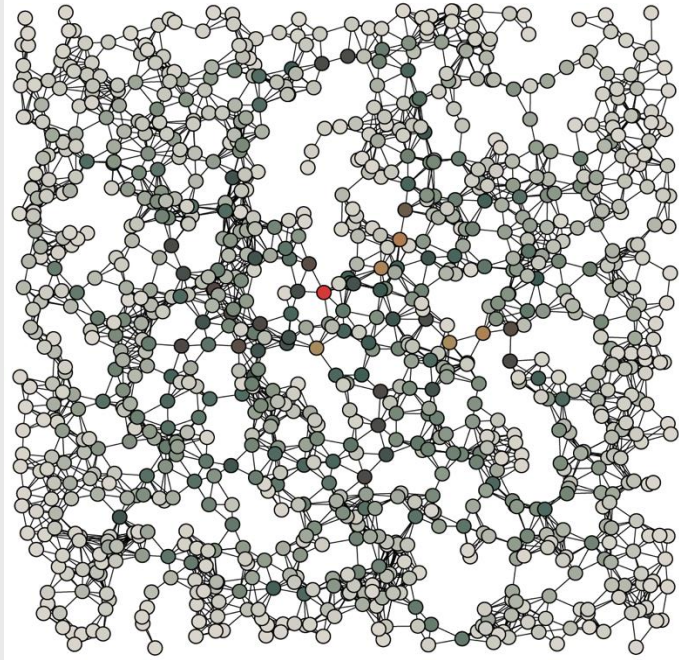
$$a_i = \alpha \sum_j A_{ij} h_j ; \quad h_i = \alpha \sum_j A_{ji} a_j$$

Hubs

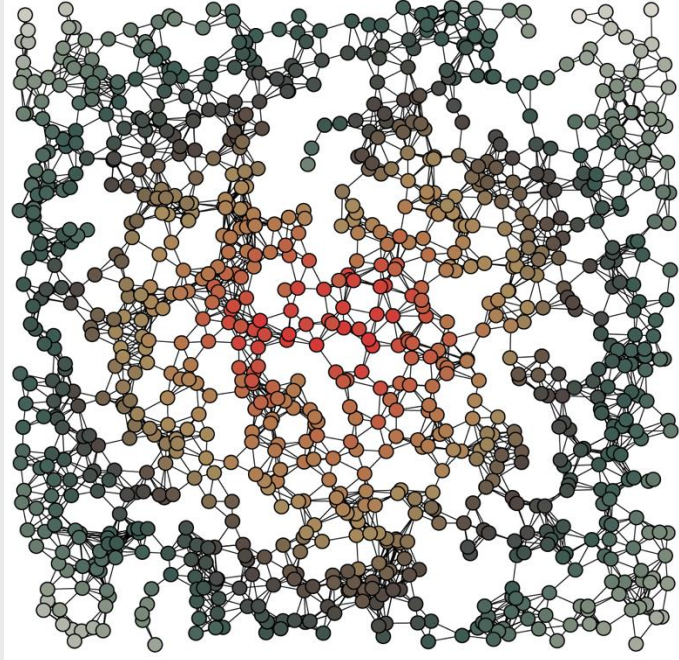


Authorities

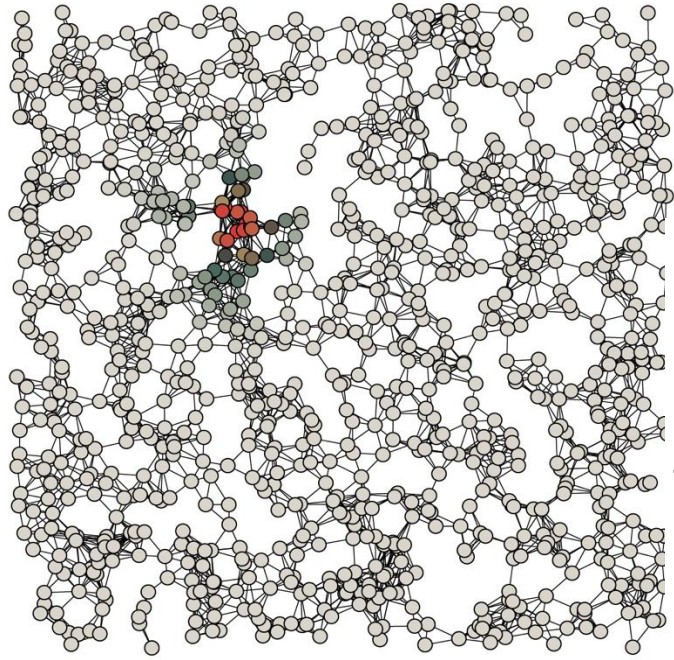




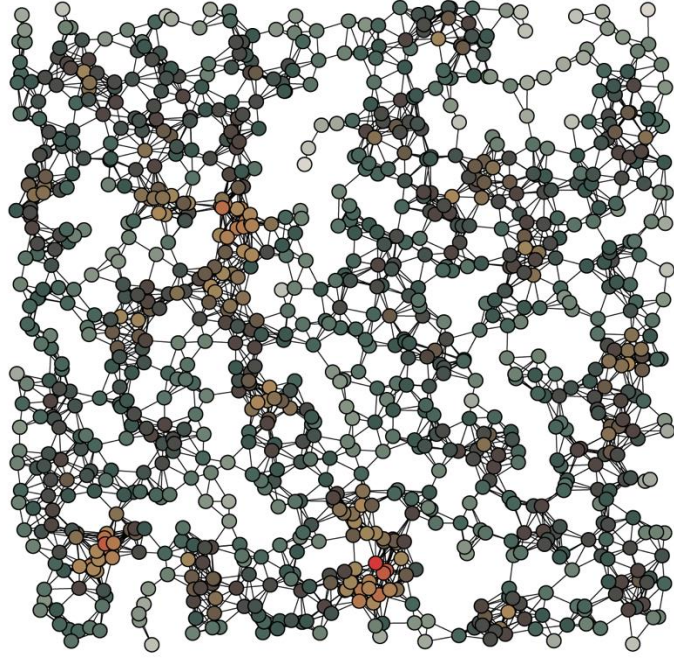
A Betweenness



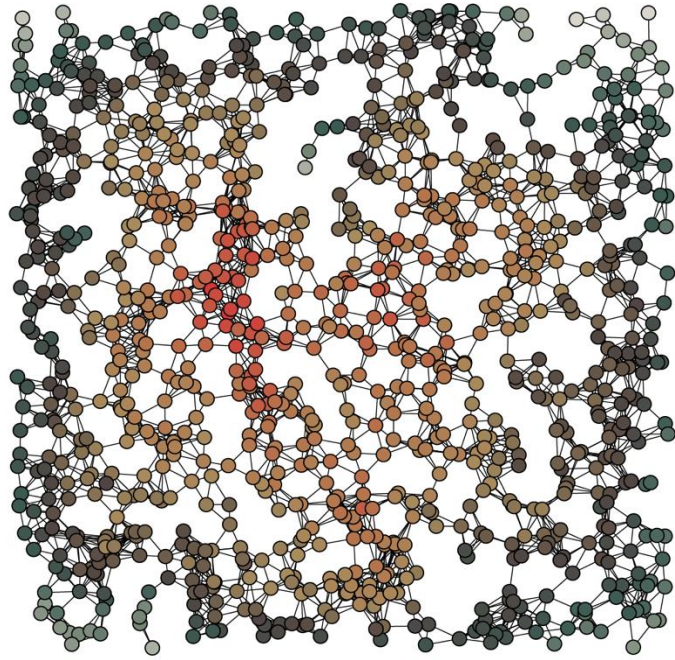
B Closeness



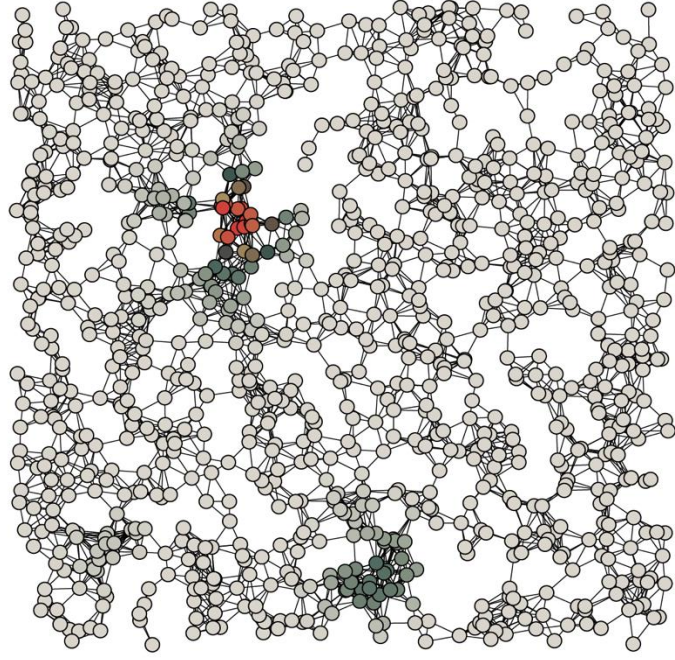
C PageRank



D Degree



E Harmonic



F Katz

Least central

Most central

Use a centrality measure that fits your theory, not the one that gives you the best results

Consider what is the real objective (e.g. is it to enable low-income individuals to increase their social capital?) (<https://petterhol.me/2019/01/11/the-importance-of-being-earnest-about-node-importance/>)

Periodic Table of Network Centrality																									
1 IA												18 VIIIA													
8000 1979	DC											518 1989	IC												
Degree	Degree															Information C									
224 1971	BC	239 2008	EBC											26 1989	kPC	275 2002	EGO	51 2004	HYPER	279 1997	AFF	399 2 001	α -C	178 1995	ECC
Betweenness	Betweenness	Endpoint BC	Endpoint BC											kPath C	kPath C	Ego	Hypergraph	Affiliation C.	α -Cent.	Eccentricity					
942 1966	CC	239 2008	PBC	3 IIIA	4 IVB	5 VB	6 VIB	7 VIIB	8 VIIIB	9 VIIIB	10 VIIIB	11 IB	12 IIB	9068 1999	HITS	573 2006	g-kPC	296 1999	GROUP	80 2006	HYPSC	34 2010	t-SC	116 1998	RAD
Closeness	Closeness	Proxy BC	Proxy BC	Hub/Authority	geodesic kPath	Groups/Classes	Hyperg. SC	296 1999	GROUP	80 2006	HYPSC	34 2010	t-Subgraph	Radiality											
1279 1972	EC	239 2008	LSBC	EBC	CBC	Δ C	MDC	EYC	CAC	EPTC	CCoef	PeC	BN	43 2009	EI	573 2006	e-kPC	573 2006	v-kPC	505 2010	WEIGHT	17 2013	TCom	116 1998	INT
Eigenvector	Eigenvector	ScaledBC	Edge BC	Commun. BC	Delta Cent.	MD Cent.	Entropy C.	Comm. Ability	Entropy PC	Clust. Coef.	PeC	Bottleneck	Essentiality I	e-disjoint kPC	v-disjoint kPC	Weighted C.	Total Comm.	Integration							
1306 1953	KS	239 2008	DBBC	RWBC	TEC	LI	MC	COMCC	ECCoef	SMD	UCC	WDC	MNC	43 2009	KL	BIP	GPI	426 1988	kRPC	116 1991	SCodd	58 2007	RWCC	586 2004	RWCC
Katz Status	Katz Status	DBounded BC	RWalk BC	Total Effects	Lobby Index	Mod. Cent.	Community C.	ECCoef	Super Mediat.	United Comp.	WDC	MNC	Clique Level	Bipartivity	GPI Power	Reachability	odid Subgraph	RWalk CC							
8053 1999	PR	239 2008	DSBC	σ	IEC	DM	LAPC	ABC	STRC	SNR	HPC	LAC	DMNC	3 2013	LR	β -C	HYP	27 2012	KEPC	13 2007	FC	0 2014	HCC		
Page Rank	Page Rank	DScaled BC	Stress	Immediate Eff.	Degree Mass	Laplacian C.	Attentive BC	Straightness C.	Silent Node R.	Harm. Prot.	Local Average	DMNC	Lurker Rank	β Cent.	Hyperbolic C.	k-edge PC	Functional C.	Hierar. CC							
484 2005	SC	613 1991	FBC	RLBC	MEC	LEVC	TC	SDC	ZC	CI	CoEWC	NC	MLC	X X	RSC	SWIPD	XXXX	0 2014	BCPR	0 2014	TPC	0 2015	EDCC		
Subgraph	Subgraph	Flow BC	RLimited BC	Mediative Eff.	Leverage Cent.	Topological C.	Sphere Degree	Zonal Cent.	Collab. Index	CoEWC	NC	Moduland C.	Resolvent SC	SWIPD	LiaComb	BCPR	Tunable PC	Effective Dist.							

Chains

Sometimes data is represented as chains

- Life trajectory: Aranda → León → Vermont → Amsterdam
- Ownership chain: (right figure)

They allow you to do other analysis:

- Importance of the node based on how often it is found in between
- Importance of the node based on how many people jump to you

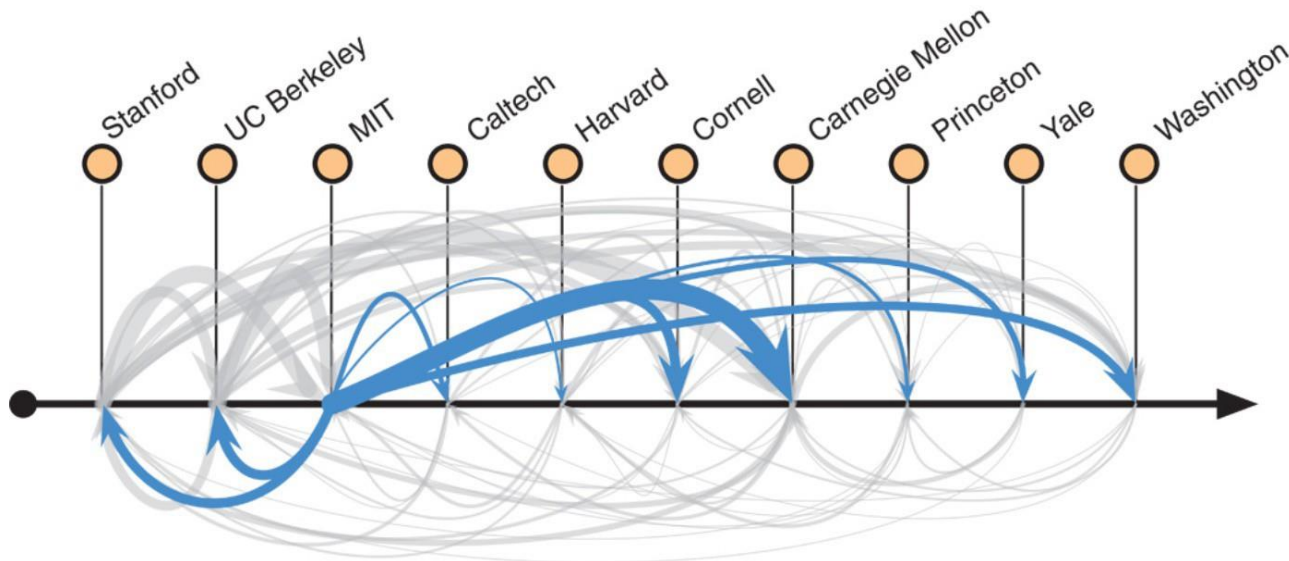


Fig. 1 Prestige hierarchies in faculty hiring networks.

Clauset, Arbesman and Larremore (2015)



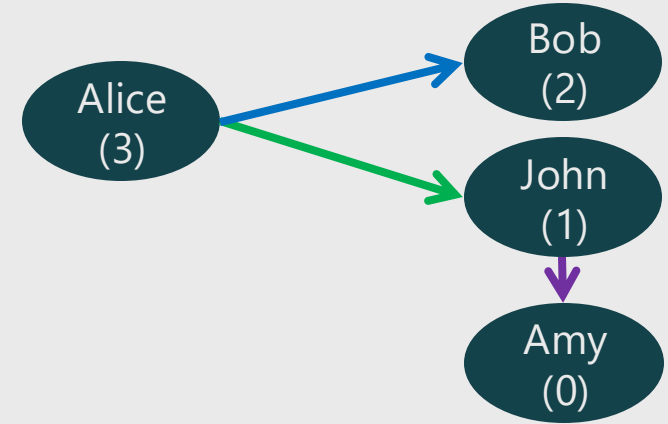
Practical 2:

Exercise 4 and 5

Linear algebra and centrality measures

Matrix multiplication: paths

Interpretation A: Presence of path between node i and j



$A^2 =$

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

@

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

=

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0	0	1
Bob	0	0	0	0
John	0	0	0	0
Amy	0	0	0	0

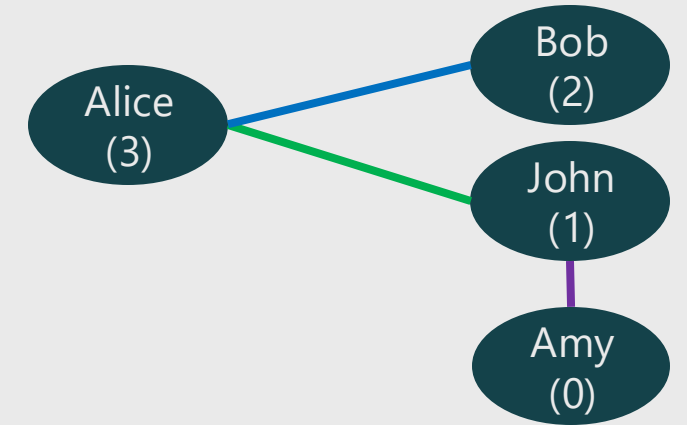
From you → to your neighbors

From your neighbors → to their neighbors

You → the neig. of your neig.

Degree

= paths of length 1



Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	1	0	0	0
John	1	0	0	1
Amy	0	0	1	0

@

Alice	1
Bob	1
John	1
Amy	1

=

	Degree
Alice	2
Bob	1
John	2
Amy	1

Eigenvector

= paths over all possible lengths

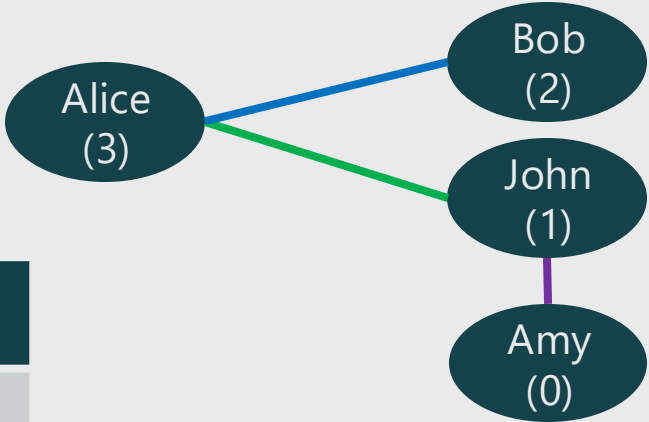
Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	1	0	0	0
John	1	0	0	1
Amy	0	0	1	0

@

Alice	1
Bob	1
John	1
Amy	1

=

	Degree
Alice	2
Bob	1
John	2
Amy	1



Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	1	0	0	0
John	1	0	0	1
Amy	0	0	1	0

@

Alice	2
Bob	1
John	2
Amy	1

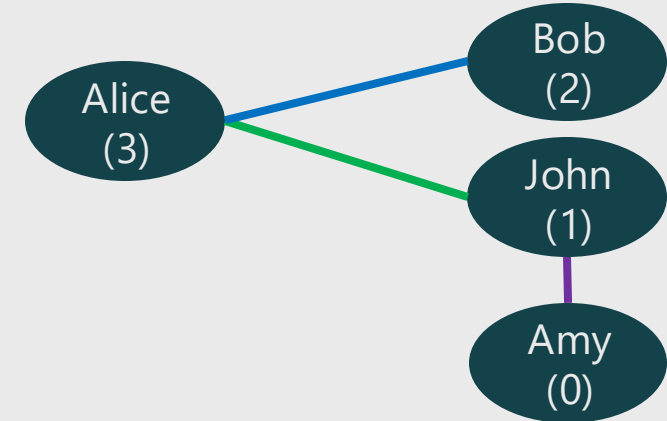
=

Alice	3
Bob	2
John	3
Amy	2

Another view on matrix multiplications: Random walks on undirected networks

Transition matrix (row-normalized A)

Target → ↓ Source	Alice	Bob	John	Amy		Target → ↓ Source	Alice	Bob	John	Amy	
Alice	0	0.5	0.5	0	@	Alice	0	0.5	0.5	0	=
Bob	1	0	0	0		Bob	1	0	0	0	
John	0.5	0	0	0.5		John	0.5	0	0	0.5	
Amy	0	0	1	0		Amy	0	0	1	0	



A random walker starting in Alice will go 50% of the times to Bob, 50% of the times to John.

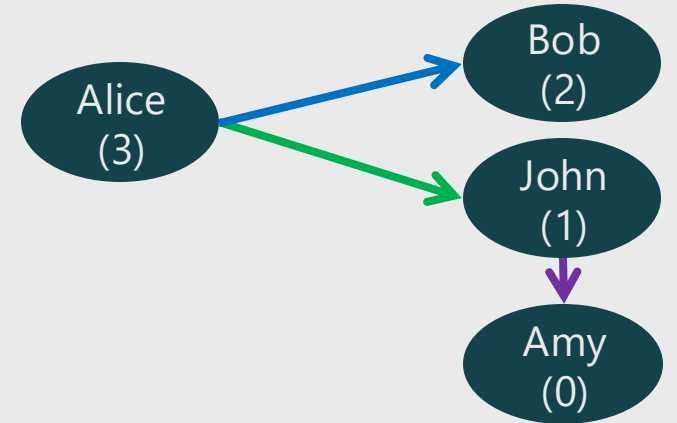
- From Bob it goes 100% of the times back to Alice
- From John it goes 50% of the times to John, 50% back to Alice

If we let the random walker walk forever → The fraction of time spend at each node converges to the **degree centrality** of the node

Another view on matrix multiplications: Random walks on directed networks

Transition matrix (row-normalized A)

Target → ↓ Source	Alice	Bob	John	Amy		Target → ↓ Source	Alice	Bob	John	Amy	
Alice	0	0.5	0.5	0	@	Alice	0	0.5	0.5	0	=
Bob	0	0	0	0		Bob	0	0	0	0	
John	0	0	0	1		John	0	0	0	1	
Amy	0	0	0	0		Amy	0	0	0	0	



A random walker starting in Alice will go 50% of the times to Bob, 50% of the times to John.

- From Bob it gets trapped
- From John it goes 100% of the times to Amy and gets trapped

If we let random walkers walk forever → They gets trapped in the extremes!

Solution: PageRank (the alpha parameter can be understood as a teletransportation probability)

Practical 3:

Working with networks using Gephi

- Follow this tutorial (slides 1–23 only!): <https://gephi.org/users/quick-start/>
- In community detection use the “stochastic blockmodel” instead of modularity maximization (or try both)
- You can choose to use our own data (<https://tinyurl.com/network-game>) or the Twitter or PPI data.

**Python exercise
notebook 2, ex.6
(if you like math)**

Recap of today

There is important information encoded in relationships/interactions

Modeling systems using networks allow us to study that information

We can represent networks using adjacencies matrixes or adjacencies lists

Network science is closely linked to linear algebra (matrix multiplication).

You can now:

- **Describe networks:** number of edges and nodes, components, density, assortativity, clustering, diameter, degree distributions.
- **Test hypothesis** using reference models.
- Find the most important nodes using **centrality measures:**
 - Random paths:
 - Degree
 - Eigenvector / PageRank / Katz / Hubs and Authorities
 - Shortest paths
 - Closeness / Harmonic: Distance to all other nodes
 - Betweenness: Presence in shortest paths between other nodes

Tomorrow:

- Network Models: generate networks and test hypothesis
- Statistical Models: REM, ERGM, SAOM